

# Scalable watermarking for identifying large language model outputs.

Dathathri et al., *Nature* (2024)

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QuantCo reading group

7 Oct 2026

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# Motivation

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**AI (artificial intelligence)**

This article is more than 2 months old

## AI labels to be compulsory on authentic-looking content under EU rules

Companies must ensure people know when they are interacting with artificially generated images, audio and text designed to look real

**Jennifer Rankin in Brussels**

Fri 31 Jul 2026 12:31 BST

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Screenshot of an AI-generated image of the French president, Emmanuel Macron, dancing in a 1980s nightclub.

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Capital at risk.

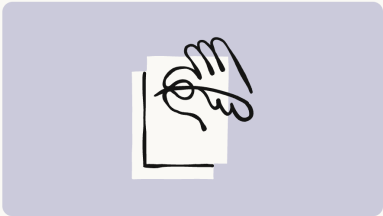
[Link.](#) EU AI Act enforcement, starting from 2nd August 2026.

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Announcements

## How Claude's text watermark works

Aug 14, 2026



**What is watermarking?**

How does watermarking affect Claude's outputs?

Which specific method of watermarking do you use?

What about cases where Claude has profited or misled human users?

Future Claude models will generate text that contains a watermark. This is a way of determining the likelihood that Claude was involved in writing the text, and we, along with several other major AI providers, are implementing this change to comply with the EU AI Act.

In this article, we share answers to some of the questions we've received about how our chosen watermarking method works, whether it affects Claude's outputs, and why we're making this change. To summarize:

[Link.](#) Anthropic adopts text watermarking.

## Writing in Acceptance Criteria



Transactions on Machine Learning Research

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9



Three things are true today:

1. Peer reviews generally put least emphasis on the criterion of clear writing. From our experience, this is true for TMLR as well.
2. Journals and conferences including TMLR are seeing a large number of submissions, including potentially AI generated or heavily AI written ones.
3. AI writing is currently often convoluted, has unnecessary jargon, and not easy to understand.

[Link](#). TMLR's call for reviewers to emphasise more on clarity of writing due to surges in AI-generated content.

# Background

# How LLMs Generate Texts

- Most LLMs generate texts using autoregressive next-token sampling.



- Example:

|                                    | Vocabulary | $p_{LM}$ | Selected token |
|------------------------------------|------------|----------|----------------|
| ... my favourite tropical fruit is | mango      | 0.50     | → lychee       |
|                                    | lychee     | 0.30     |                |
|                                    | papaya     | 0.15     |                |
|                                    | durian     | 0.05     |                |



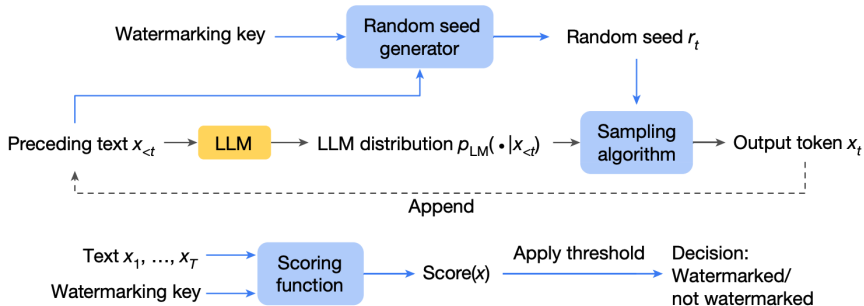
# Existing Approaches for LLM Text Detection

- Retrieval-based detection
  - ▶ Keeping a growing record of all generated texts and checking against it for matches.
  - ▶ Requires scale and coordination, and raises privacy concerns.
- Post hoc detection
  - ▶ Using the statistical features of text or training a ML-based classifier.
  - ▶ Computationally expensive; inconsistent performance (especially out-of-domain).
- Watermarking
  - ▶ Marking the generated text in some way so that it can subsequently be identified.
  - ▶ Three approaches:
    - ★ During generation. → This paper.
    - ★ Editing already generated text.
    - ★ Altering LLM training data.
  - ▶ Often involves a quality (of generated texts) and effect (of detection) trade-off.

[XL: use plots?]

## SynthID-Text

# SynthID-Text overview



- Non-distortionary.
- Computationally efficient.

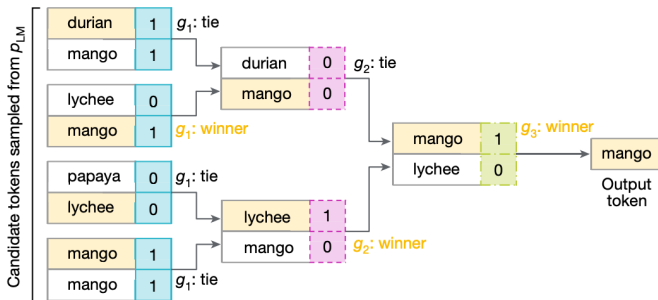
Fig. 1, Dathathri et al.

# Watermarking – Tournament Sampling

General principle: select a token from the LLM distribution that is likely to score higher under some random watermarking functions.

**Tournament Sampling:** achieve the above using a tournament-like process.

**Watermarking function:**  $g_l(x_t, r_t)$ , where  $x_t$  is a candidate token, and  $r_t$  is a random seed.



Larger  $m \rightarrow$  better detectability, but it does not increase indefinitely. They use  $m = 30$ .

$$\text{Score}(x) = \frac{1}{mT} \sum_{t=1}^T \sum_{l=1}^m g_l(x_t, r_t) .$$

Intuition:  $\text{Score}(x)$  takes higher values for a watermarked sequence.

- Example:  $g_l(x, r) = \mathbb{1}\{u_l(x, r) > 0.5\}$ , where  $u_l(x, r)$  is a pseudo-random function generating values in  $[0, 1]$ .
- In practice,

# A frequentist approach to detection

[XL: talk about hypothesis testing]

**Definition 16** (Single-token non-distortionary sampling algorithm). *A sampling algorithm  $\mathcal{S} : \Delta V \times \mathcal{R} \rightarrow V$  is (single-token) non-distortionary if for any probability distribution  $p \in \Delta V$  and token  $x \in V$ :*

$$\mathbb{E}_{r \sim \text{Unif}(\mathcal{R})} [\mathbb{P}(\mathcal{S}(p, r) = x)] = p(x).$$

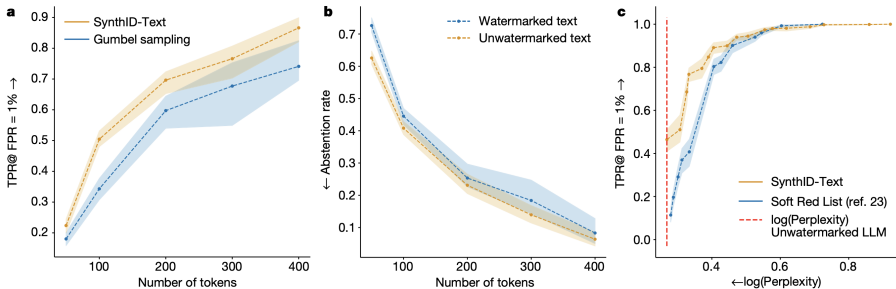
Notation:

- $V$  = set of candidate tokens (vocabulary).
- $\Delta V$  = probability simplex over  $V = \{p : V \rightarrow [0, 1] \mid \sum_{x \in V} p(x) = 1\}$ .
- $\mathcal{R}$  = set of possible random seeds.

[XL: highlight box] In words: A sampling algorithm is non-distortionary if in expectation the distribution on the next token is identical before and after watermarking.

**Strong non-distortion:** the above is extended for **sequences** of text.

# Empirical evaluation



**Fig.3** Detection performance of SynthID-Text.



## Computational complexity

| Sampling method                        | Latency (ms/token) | Increase (%) |
|----------------------------------------|--------------------|--------------|
| Baseline                               | 15.527             | –            |
| Tournament sampling ( $m = 30$ layers) | 15.615             | 0.57         |
| Gumbel sampling                        | –                  | 0.26         |
| Soft Red List                          | –                  | 0.28         |

Table: Gemma 7B-IT latency on 4 v5e TPUs.

SynthID-Text can be integrated into large-scale production systems with “negligible additional computational overhead”.

# Some thoughts